Solution The curves in Fig.1.18 (a) and Fig.1.18 (c) do not represent a function as the vertical lines meet the curves in two points P and Q.

The curves in Fig.1.18 (b) and Fig.1.18 (d) represent a function as the vertical lines meet the curve in at most one point.

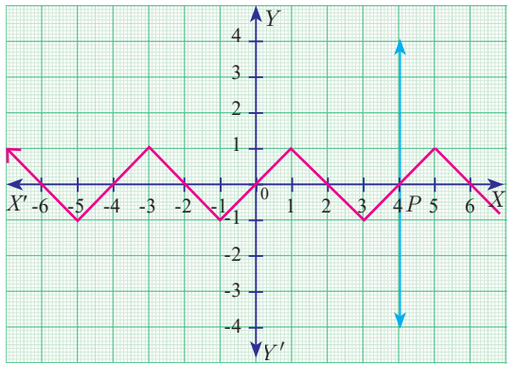


Fig. 1.18(d)

Note

Any equation represented in a graph is usually called a curve.

Example 1.11 Let $A = \{1,2,3,4\}$ and $B = \{2,5,8,11,14\}$ be two sets. Let $f : A \to B$ be a function given by $f(x) = 3x - 1$. Represent this function

(i) by arrow diagram (ii) in a table form

(iii) as a set of ordered pairs (iv) in a graphical form

Solution

$A = \{1,2,3,4\}$; $B = \{2,5,8,11,14\}$; $f(x) = 3x - 1$

$f(1) = 3(1) - 1 = 3 - 1 = 2$; $f(2) = 3(2) - 1 = 6 - 1 = 5$

$f(3) = 3(3) - 1 = 9 - 1 = 8$; $f(4) = 4(3) - 1 = 12 - 1 = 11$

(i) Arrow diagram

Let us represent the function $f : A \to B$ by an arrow diagram (Fig.1.19).

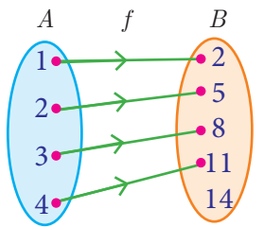


Fig. 1.19

(ii) Table form

The given function f can be represented in a tabular form as given below

x	1	2	3	4
$f(x)$	2	5	8	11

(iii) Set of ordered pairs

The function f can be represented as a set of ordered pairs as

$f = \{(1,2),(2,5),(3,8),(4,11)\}$

(iv) Graphical form

In the adjacent XY-plane the points (1,2), (2,5), (3,8), (4,11) are plotted (Fig.1.20).

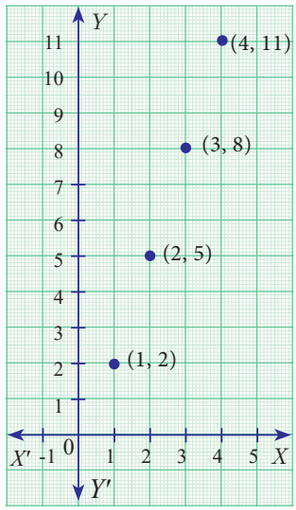


Fig. 1.20

10th_Maths_Chapter 1_English.indd 16 12/27/2024 3:08:50 PM